



NETCRADUS PVT. LTD.

Securing Businesses with
AI-Driven Cyber Intelligence



MANAGED SOC
24/7 Monitoring



**CLOUD
SECURITY**



**VAPT
SERVICES**



**THREAT DETECTION
& RESPONSE**



**AI-POWERED
SIEM**

CONNECT WITH US



WEBSITE
www.netcradus.in



TOLLFREE NO.
1800121008800



EMAIL
info@netcradus.com

INDIA | UK | GLOBAL SERVICES



01 · OVERVIEW

EXECUTIVE SUMMARY

The modern enterprise threat landscape has been permanently altered by the convergence of AI-generated malware, autonomous attack toolkits, supply-chain infiltration, and state-sponsored persistent threats operating at machine speed. Traditional Security Operations Centres, built on rule-based SIEM correlation and human analyst triage, are structurally incapable of responding within the sub-minute timelines that advanced adversaries now operate within.

The Autonomous Cyber Immune System (ACIS) is a next-generation, bio-inspired security platform that replaces reactive, human-latency-bound defence with a continuously operating, self-learning, closed-loop immune system. Modelled on the adaptive properties of human biological immunity — self/non-self discrimination, proportionate response, and immunological memory — ACIS autonomously detects, analyses, contains, remediates, and learns from every threat event across the entire enterprise estate.

This white paper presents the full ACIS technical architecture, structured around five purpose-engineered defence layers: an AI-Powered SIEM for intelligent log ingestion and detection; an Autonomous SOAR engine for sub-200ms orchestrated response; an AI Red Team Simulator for continuous adversarial validation; a Threat Intelligence Swarm for collective learning; and a Self-Healing and Deception Engine for autonomous remediation and adversary misdirection. Together, these layers form a Closed-Loop Defence Mode (CLDM) that eliminates the human bottleneck from every critical phase of the security operations lifecycle.

KEY PLATFORM CAPABILITIES

- Five discrete, interdependent defence layers — no single point of failure, no architectural gaps
- Closed-Loop Defence Mode (CLDM): detect → analyse → contain → remediate → learn in < 200 ms
- AI-Powered SIEM: 10M+ events/second ingestion with sub-10ms detection latency
- Autonomous SOAR: 240+ playbooks, 140+ native integrations, transactional rollback
- AI Red Team Simulator: continuous 24/7 adversarial testing against live production controls
- Threat Intelligence Swarm: federated collective learning across the ACIS customer fleet
- Self-Healing Engine: automated endpoint remediation, patch deployment, and deception layer activation

02 · CONTEXT

1. THE MODERN THREAT LANDSCAPE

Contemporary adversaries operate with automation, artificial intelligence, and commoditised toolkits that compress the full attack lifecycle — from initial access to data exfiltration — to under 62 minutes (CrowdStrike Global Threat Report 2024). This velocity renders human-in-the-loop security architectures operationally obsolete for primary threat containment. ACIS is designed specifically for this environment.

1.1 Structural Drivers of Security Failure

- AI-generated polymorphic malware mutates binary signatures on every execution cycle, defeating signature-based AV and traditional SIEM correlation rules entirely.



- ▶ Mean time from initial access to lateral movement now averages 62 minutes, with the fastest recorded intrusion (2024) completing lateral movement in under 2 minutes and 7 seconds.
- ▶ Global cybersecurity analyst shortage exceeds 4 million professionals (ISC² 2024), creating systemic alert fatigue — 45% of SOC alerts are never investigated.
- ▶ Supply-chain attacks increased 742% between 2019 and 2024, targeting the software build pipeline where traditional endpoint controls have no visibility.
- ▶ Ransomware-as-a-Service ecosystems have reduced the technical barrier to entry for sophisticated attacks to near-zero, democratising nation-state-grade TTPs.

1.2 The ACIS Response

ACIS addresses these structural failures through five complementary autonomous layers that collectively eliminate the three root causes of security breach dwell time: detection latency, analyst bottleneck, and remediation lag. The Closed-Loop Defence Mode operates continuously, autonomously, and in parallel across all five layers — with no dependency on human reaction time for primary threat containment.

03 · ARCHITECTURE

2. ACIS DESIGN PHILOSOPHY: THE CYBER IMMUNE SYSTEM

ACIS draws its architectural principles directly from biological immunology. The human immune system achieves something that no conventional security architecture has matched: continuous, autonomous, adaptive defence with perfect self/non-self discrimination, proportionate multi-layer response, and persistent immunological memory. ACIS maps each biological mechanism to a concrete technical implementation.

BIOLOGICAL MECHANISM	CYBER ANALOGUE	ACIS IMPLEMENTATION
Self / Non-Self Discrimination	Behavioural Baselining	Per-entity DNA profiling via UEBA — Layer 1 AI-SIEM
Innate Immunity	Signature & Rule Detection	YARA-X, Sigma rules, IOC matching — Layer 1 AI-SIEM
Adaptive Immunity	AI Threat Classification	GNN + XGBoost ensemble, continuous retraining — Layer 1 & 3
Inflammatory Response	Automated Containment	Sub-200ms SOAR playbook execution — Layer 2
Immune Memory	Threat Intelligence Swarm	Persistent adversary knowledge graph — Layer 3
Tissue Repair	Self-Healing Engine	Automated rollback, patch, deception activation — Layer 4
Stress Testing	Red Team Simulation	Continuous BAS and adversarial emulation — Layer 5

3. FIVE-LAYER CLOSED-LOOP DEFENCE ARCHITECTURE

ACIS implements defence-in-depth through five purpose-engineered, interdependent layers. Each layer operates independently and contributes unique defensive capability, but their power is multiplied through tight orchestration via the CLDM engine. A threat event detected in any layer triggers coordinated defensive escalation across all others.

#	LAYER NAME	PRIMARY FUNCTION
1	AI-Powered SIEM	Unified log ingestion, behavioural baselining, and AI-driven threat detection across all telemetry sources
2	Autonomous SOAR	Sub-200ms orchestrated response: playbook execution, containment, network isolation, and identity plane actions
3	Threat Intelligence Swarm	Federated collective intelligence: multi-source IOC ingestion, adversary knowledge graph, and fleet-wide learning
4	Self-Healing & Deception Engine	Automated endpoint remediation, patch orchestration, deception layer deployment, and adversary misdirection
5	AI Red Team Simulator	Continuous 24/7 adversarial emulation, breach-and-attack simulation, and autonomous control validation

4. LAYER 1 — AI-POWERED SIEM

LAYER 1 · AI-POWERED SIEM — LOG INGESTION & DETECTION

The Intelligent Sensory Layer — Unified Telemetry, Behavioural Baselining & AI-Driven Threat Detection

- ◆ Unified telemetry ingestion from endpoints, network, cloud, identity, OT/IoT, and application layers at 10M+ events/second
- ◆ Per-entity behavioural DNA profiling: statistical baselines built per user, device, process, and service account over rolling 90-day windows
- ◆ AI-native detection: Isolation Forest + DBSCAN anomaly detection running inline on the streaming pipeline — zero-day capable without signatures
- ◆ YARA-X and Sigma rule engine: hardware-accelerated multi-pattern matching against 8.4M+ IOC signatures updated in real-time from the Threat Swarm
- ◆ UEBA: lateral movement, credential stuffing, insider threat, and privilege escalation detection
- ◆ MITRE ATT&CK automatic mapping: every detected event tagged to ATT&CK tactic, technique, and sub-technique within the stream processor
- ◆ Unified risk scoring engine: per-entity composite risk score propagated in real-time to Layer 2 SOAR for response prioritisation
- ◆ Cloud-native telemetry: native connectors for AWS CloudTrail, Azure Monitor, GCP Audit Logs, Kubernetes audit logs, and all major SaaS APIs

4.1 AI-SIEM Technology Stack

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Event Streaming	Apache Kafka 3.x (multi-AZ, exactly-once)	Durable ordered event bus at 10M+ msg/sec; consumer groups per detection engine
Stream Processor	Apache Flink 1.18 (stateful CEP)	Sub-10ms windowed aggregation, complex event processing, and rule evaluation
Endpoint Agent	Rust kernel driver + eBPF hooks	Zero-overhead syscall interception; file, network, and process telemetry at ring-0
Network Capture	libpcap / AF_PACKET + Intel DPDK	Kernel-bypass line-rate packet capture; L4–L7 deep packet inspection
Log Aggregation	Fluent Bit 3.x → OpenSearch 2.x	Lightweight log forwarding, normalisation, and centralised search with retention tiers
Signature Engine	Hyperscan (Intel) + YARA-X	Hardware-accelerated multi-pattern regex matching at memory bandwidth limits
Anomaly Detection	Isolation Forest + DBSCAN (scikit-learn/Ray)	Distributed unsupervised ML anomaly scoring across all entity streams in real-time
UEBA Engine	PyTorch LSTM + statistical peer-group models	Temporal behavioural modelling with peer-group deviation scoring

ACIS · AUTONOMOUS CYBER IMMUNE SYSTEM		
COMPONENT	TECHNOLOGY / STACK	PURPOSE
Schema Registry	Apache Avro + Confluent Schema Registry	Typed, versioned telemetry schemas ensuring cross-source normalisation
Cloud Collector	ACIS Cloud Collector (Go, multi-cloud)	Unified ingestion from AWS, Azure, GCP, and SaaS APIs via ACIS Collector agent

4.2 Detection Performance

ATTRIBUTE	SPECIFICATION
Ingestion Throughput	10,000,000+ events per second (horizontally scaled)
Detection Latency (P99)	< 10 milliseconds from event arrival to risk score
IOC Signature Library	8.4 million+ signatures; real-time updates from Threat Swarm
Anomaly Detection Accuracy	> 99.3% precision at production throughput
False Positive Rate	< 0.7% (adversarial training hardened)
MITRE ATT&CK Coverage	100% of Enterprise ATT&CK v14 techniques mapped
Supported Log Sources	850+ native parsers; 1,200+ via OpenTelemetry collector
Retention: Hot / Warm / Cold	24 hours / 90 days / 7 years (Apache Iceberg on object store)

5. LAYER 2 — AUTONOMOUS SOAR

LAYER 2 · AUTONOMOUS SOAR — ORCHESTRATE & RESPOND

The Autonomous Response Engine — Sub-200ms Closed-Loop Threat Containment & Neutralisation

◆	Sub-200ms response latency from threat confirmation to first containment action — 3,600× faster than the median human SOC response time
◆	240+ pre-built automated response playbooks covering ransomware, APT lateral movement, insider threat, DDoS, supply-chain compromise, and BEC
◆	Graduated response model: monitor → alert → contain → isolate → neutralise → eradicate — proportionate to threat confidence score
◆	Network micro-isolation: SDN policy push to quarantine compromised hosts at the switch level within 500ms, without requiring agent-side action
◆	Identity plane response: autonomous revocation of OAuth tokens, MFA enforcement, session termination, and account lockdown via IdP API integrations
◆	Cloud workload response: ECS task termination, EC2 instance isolation via Security Group mutation, AKS pod eviction, and GCP instance snapshot-and-terminate
◆	Atomic execution with transactional rollback: if any response action fails mid-execution, all partial changes revert automatically
◆	Configurable autonomy thresholds: organisations set the confidence score above which CLDM operates fully autonomously (default: 95%)

5.1 SOAR Technology Stack

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Orchestration Engine	Custom ACIS SOAR core (Go microservices)	Sub-200ms playbook runtime with transactional rollback and atomic execution
Playbook Runtime	ACIS Playbook DSL (YAML) + OPA Rego policies	Human-readable response playbooks with policy-as-code constraint evaluation
Response Decision	Bayesian ensemble + policy matrix engine	Threat type × confidence × asset criticality → playbook selection in < 5ms
API Gateway	Kong Enterprise + OAuth 2.0 + mTLS	Authenticated, rate-limited API surface for all 140+ security control integrations
Identity Response	Okta Workflows API / Azure AD Graph API / Ping	Programmatic session termination, MFA enforcement, and OAuth token revocation
Network Control	OpenFlow / NETCONF + Cisco / Palo Alto APIs	SDN micro-segmentation and firewall rule push across hybrid network fabric
Cloud Orchestration	AWS SDK v3 / Azure SDK / GCP Client Libraries	Cross-cloud workload isolation, snapshot, and containment via native cloud APIs

ACIS · AUTONOMOUS CYBER IMMUNE SYSTEM		
COMPONENT	TECHNOLOGY / STACK	PURPOSE
Audit Ledger	Apache Kafka + Immutable OpenSearch index	Tamper-evident, append-only log of every autonomous action with full provenance
Integration Bus	Apache Camel + Kafka Connect	140+ certified connectors to EDR, NGFW, IdP, ITSM, SIEM, and threat intel platforms
Workflow State	Apache Cassandra 5.x (multi-region active-active)	Sub-millisecond workflow state read/write for parallel containment stream coordination

5.2 CLDM — Closed-Loop Defence Mode: Six-Phase Execution

The Closed-Loop Defence Mode is the operational core of ACIS — the engine that transforms passive detection into autonomous remediation within a single, continuous feedback cycle.

- ▶ Phase 1 — Sense: Multi-plane telemetry collected from EDR agents, network taps, cloud APIs, and identity providers and streamed to the AI-SIEM layer.
- ▶ Phase 2 — Analyse: Sub-10ms threat classification via Flink CEP rules and ML ensemble; MITRE ATT&CK mapping and attack chain graph reconstruction within 500ms.
- ▶ Phase 3 — Decide: Response decision matrix selects playbook based on threat type, confidence score, asset criticality, and regulatory constraint validation.
- ▶ Phase 4 — Act: Atomic parallel execution of containment actions across network, identity, endpoint, and cloud planes with transactional rollback.
- ▶ Phase 5 — Verify: Post-action telemetry re-analysed within 30 seconds to confirm threat neutralisation; escalation to next severity tier if indicators persist.
- ▶ Phase 6 — Learn: Confirmed threat artifacts promoted to detection rules, ML models fine-tuned, and intelligence published to the Threat Intelligence Swarm.

5.3 Response Metrics

ATTRIBUTE	SPECIFICATION
CLDM Response Latency (P99)	< 200 milliseconds (detection to first containment action)
Pre-built Response Playbooks	240+ covering all MITRE ATT&CK tactic categories
Native Security Integrations	140+ certified connectors (EDR, NGFW, IdP, ITSM, SIEM, cloud)
Parallel Containment Streams	Unlimited concurrent; network, identity, endpoint, and cloud planes
Rollback Guarantee	100% atomic — no orphaned containment states on execution failure
Human Override Latency	< 2 seconds from SOC analyst decision to playbook abort/modify
Incident Ticket Creation	Automated ServiceNow / Jira ticket with full context in < 5 seconds

6. LAYER 3 — THREAT INTELLIGENCE SWARM

LAYER 3 · THREAT INTELLIGENCE SWARM — COLLECTIVE LEARNING

The Immunological Memory Layer — Federated Intelligence, Adversary Knowledge Graph & Fleet-Wide Collective Defence

- ◆ Federated learning architecture: anonymised threat intelligence shared across the entire ACIS customer fleet — every organisation's detection improves every other's
- ◆ Multi-source IOC ingestion: STIX/TAXII 2.1 feeds, ISACs, dark web telemetry, vendor threat APIs (Mandiant, Recorded Future, CrowdStrike Intel)
- ◆ Adversary knowledge graph: persistent property graph database mapping threat actor TTPs, infrastructure clusters, campaign relationships, and victimology
- ◆ Graph Neural Network (GNN) correlation: GraphSAGE model traverses entity relationships to infer attack paths and predict next-stage adversary actions
- ◆ LLM-powered threat narratives: fine-tuned large language model generates human-readable attack summaries and executive briefings within 30 seconds of detection
- ◆ Threat actor fingerprinting: adversary infrastructure clustering via passive DNS correlation, TLS certificate pivoting, and behavioural TTPs
- ◆ Collective blocking: IOC confirmed malicious by any ACIS customer fleet member is propagated to all Layer 1 detection engines within < 60 seconds
- ◆ Explainable AI (XAI) module: every threat classification includes a human-interpretable explanation of the evidence chain — reducing analyst validation time by 94%

6.1 Threat Intelligence Swarm Technology Stack

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Adversary Knowledge Graph	JanusGraph + Apache Cassandra backend	Property graph with TTL-based IOC lifecycle management and adversary TTP mapping
Graph Neural Network	PyTorch Geometric (GraphSAGE + GAT)	Attack path inference, entity relationship scoring, and lateral movement prediction
Federated Learning	Flower (flwr) 1.x framework	Privacy-preserving collaborative model training across ACIS fleet — no raw data sharing
Intel Feed Ingestion	STIX/TAXII 2.1 clients + custom adapters	Normalised IOC ingestion from 60+ threat intelligence providers and ISAC feeds
IOC Storage & Search	Elasticsearch 8.x (ILM-managed)	Full-text search and aggregation across 8.4M+ signatures with sub-5ms query latency
LLM Threat Narratives	Fine-tuned Llama 3 70B (private deployment)	On-premises LLM inference for threat summaries — no data leaves customer boundary
NLP Inference Serving	NVIDIA Triton Inference Server + TensorRT	GPU-accelerated model inference at < 5ms P99 latency for production throughput

ACIS · AUTONOMOUS CYBER IMMUNE SYSTEM		
COMPONENT	TECHNOLOGY / STACK	PURPOSE
Collective IOC Bus	Apache Kafka (dedicated swarm cluster)	< 60-second IOC propagation across fleet with cryptographic provenance attestation
Dark Web Telemetry	ACIS DarkSensor (Tor + I2P passive collector)	Passive dark web monitoring for credential exposure, actor chatter, and 0-day trading
XAI Engine	SHAP (SHapley Additive exPlanations) + LIME	Per-prediction explainability — evidence chain surfaced in ACIS analyst interface

6.2 Intelligence Metrics

ATTRIBUTE	SPECIFICATION
IOC Propagation Latency	< 60 seconds (confirmed IOC → fleet-wide blocking)
Active Threat Intelligence Feeds	60+ providers: ISACs, Mandiant, Recorded Future, VirusTotal, AlienVault OTX
Adversary Knowledge Graph Size	140M+ nodes; 890M+ edges (adversary TTPs, infra, campaigns)
Fleet Collective Learning	Anonymised intelligence from entire ACIS customer base
Threat Narrative Generation	< 30 seconds from detection to executive-ready report
XAI Explanation Coverage	100% of ML classifications include human-interpretable evidence
Dark Web Coverage	Tor, I2P, and Telegram threat channel passive monitoring

7. LAYER 4 — SELF-HEALING & DECEPTION ENGINE

LAYER 4 · SELF-HEALING & DECEPTION ENGINE

The Tissue-Repair & Adversary Misdirection Layer — Autonomous Remediation, Patch Orchestration & Active Deception

◆	Automated endpoint remediation: malicious processes terminated, registry keys reverted, file system changes rolled back via VSS/snapshot, and persistence mechanisms removed
◆	Vulnerability patch orchestration: ACIS identifies the CVE exploited, retrieves the validated vendor patch, and deploys via rolling zero-downtime update within minutes
◆	System integrity restoration: cryptographic verification of critical OS binaries against known-good hashes; automatic restoration of tampered files from a tamper-evident integrity store
◆	Active deception layer: on detection of active reconnaissance, CLDM activates synthetic honeypot assets — fake servers, credentials, file shares, and network segments
◆	High-interaction honeypot telemetry: adversary interactions with deception assets are deeply logged, providing ground-truth adversary TTP data fed directly to the Threat Intelligence Swarm
◆	Canary token deployment: ACIS auto-deploys canary tokens (URLs, documents, AWS keys, DNS) across the estate; any interaction immediately confirms active compromise
◆	Security posture scoring: real-time CVSS 4.0-adjusted risk score per asset, per business unit, and organisation-wide
◆	Regulatory remediation automation: post-incident compliance reports auto-generated for GDPR Article 33, PCI-DSS 12.10, and ISO 27001 Annex A.16 within minutes of incident closure

7.1 Self-Healing & Deception Technology Stack

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Endpoint Remediation	ACIS Remediation Agent (Rust) + OS VSS/APFS snapshots	File-level rollback, process termination, and registry restoration with atomic commit
Patch Orchestration	ACIS Patch Engine + vendor NVD/OVAL feeds + Ansible	CVE-to-patch mapping, vendor retrieval, and zero-downtime rolling deployment
Integrity Verification	SHA-3 binary manifest + ACIS Integrity Ledger (Cassandra)	Continuous cryptographic verification of OS and application binaries
Deception Platform	OpenCanary + ACIS Deception Fabric (Go)	High-interaction honeypots: fake SMB, SSH, RDP, HTTP, and database services
Canary Tokens	Canarytokens.org engine (self-hosted) + ACIS integration	URL, document, DNS, AWS key, and email canaries deployed across estate
Honeypot Telemetry	Dedicated Kafka topic → Threat Intelligence Swarm	Real-time adversary interaction data enriching the ACIS adversary knowledge graph

ACIS · AUTONOMOUS CYBER IMMUNE SYSTEM		
COMPONENT	TECHNOLOGY / STACK	PURPOSE
Posture Scoring	CVSS 4.0 engine + ACIS Asset Criticality Model	Continuous per-asset risk scoring incorporating exploit availability and exposure
Compliance Automation	ACIS Compliance Engine + Jinja2 report templates	Auto-generated GDPR Art.33, PCI-DSS, and ISO 27001 post-incident reports
Secrets Rotation	HashiCorp Vault Enterprise (dynamic secrets)	Automatic credential rotation for compromised service accounts and API keys

8. LAYER 5 — AI RED TEAM SIMULATOR

LAYER 5 · AI RED TEAM SIMULATOR — CONTINUOUS ATTACK EMULATION

The Adversarial Validation Layer — 24/7 Autonomous Breach Simulation, Control Testing & Resilience Assurance

- ◆ Continuous 24/7 adversarial emulation against live production security controls — not periodic snapshots, but a persistent autonomous red team operating at all times
- ◆ AI-driven attack scenario generation: LLM-assisted threat modelling generates novel attack chains based on current estate topology, detected vulnerabilities, and MITRE ATT&CK kill-chains
- ◆ Breach-and-Attack Simulation (BAS): automated execution of 1,400+ attack scenarios spanning initial access, execution, persistence, privilege escalation, lateral movement, and exfiltration
- ◆ MITRE ATT&CK coverage mapping: post-simulation gap analysis identifies which techniques were not detected or blocked — directly feeding Layer 1 detection rule improvement
- ◆ Purple team automation: AI Red Team shares attack telemetry in real-time with Layer 1 AI-SIEM — enabling live validation of detection efficacy during simulated attacks
- ◆ Assumed breach simulation: exercises conducted from inside the network boundary, simulating post-initial-access adversary behaviour to test lateral movement and exfiltration controls
- ◆ Cloud attack emulation: native simulation of AWS, Azure, and GCP attack paths including IAM privilege escalation, S3 data exfiltration, metadata service exploitation
- ◆ Simulation results dashboard: CISO-level visualisation of control coverage, detection gaps, mean time to detect (MTTD), and remediation velocity — updated continuously

8.1 AI Red Team Simulator Technology Stack

COMPONENT	TECHNOLOGY / STACK	PURPOSE
BAS Engine	ACIS Attack Emulator (Go) + Caldera 5.x (MITRE)	MITRE ATT&CK-aligned attack scenario execution with 1,400+ techniques
AI Scenario Generator	Fine-tuned LLM (Llama 3 70B) + ATT&CK Navigator	Estate-aware novel attack chain generation and threat modelling
Network Emulation	ACIS Network Simulator + Mininet	Virtual network topology emulation for non-disruptive production-safe testing
Cloud Attack Emulation	Pacu (AWS) + Stormspotter (Azure) + GCP attack suite	Native cloud attack path simulation across all three hyperscale providers
Purple Team Telemetry	Dedicated Kafka topic → Layer 1 AI-SIEM (real-time)	Live attack telemetry shared to detection engine for concurrent validation
Coverage Mapping	ATT&CK Navigator + ACIS Gap Analysis Engine	Automated post-simulation detection gap identification and rule recommendation
Reporting Engine	ACIS Report Generator (Jinja2 + Python)	Automated CISO dashboards, board-level summaries, and technical gap reports

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Scheduling	Kubernetes CronJob + ACIS Campaign Orchestrator	Continuous, randomised simulation scheduling to prevent detection pattern learning

8.2 Red Team Simulator Metrics

ATTRIBUTE	SPECIFICATION
Attack Scenarios Library	1,400+ across all MITRE ATT&CK Enterprise technique categories
Simulation Frequency	Continuous 24/7; randomised scheduling prevents defender pattern learning
ATT&CK Framework Coverage	100% of Enterprise ATT&CK v14 tactics; 97% of techniques
Cloud Platform Coverage	AWS, Azure, GCP — native attack path emulation for all three
Purple Team Telemetry Latency	Real-time attack data shared to detection engine during simulation
Gap Report Generation	Automated post-simulation gap report available within 5 minutes
Mean Time to Detect Tracking	MTTD tracked per technique, per layer, and per estate segment
Production Safety	All simulations safety-bounded — no payload detonation on live assets

9. CONSOLIDATED ENTERPRISE TECHNOLOGY STACK

The following sections provide a consolidated reference of the full ACIS technology stack across all platform domains, enabling enterprise architects and security engineering teams to assess integration requirements, technology compatibility, and deployment prerequisites.

9.1 Infrastructure & Platform

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Container Runtime	Kubernetes 1.29 + CRI-O	Production container orchestration with ACIS security policies via OPA Gatekeeper
Service Mesh	Istio 1.20 (mTLS everywhere)	Zero-trust inter-service communication; automatic mutual TLS certificate rotation
Infrastructure-as-Code	Terraform + Helm 3 + Kustomize	Reproducible, version-controlled infrastructure across all cloud providers
GitOps	GitLab CI + ArgoCD	Automated build, security scan (SAST/DAST/SCA), and zero-downtime deployment
Observability	OpenTelemetry → Grafana + Tempo + Loki	Unified metrics, traces, and logs for ACIS platform health and SLA monitoring
Multi-Cloud	AWS + Azure + GCP (active-active)	No vendor lock-in; cross-cloud active-active for 99.999% platform availability
HSM	AWS CloudHSM / Thales Luna (FIPS 140-2 L3)	Hardware-secured cryptographic key operations for all ACIS data encryption
Secrets Management	HashiCorp Vault Enterprise	Dynamic secrets, PKI management, and encryption-as-a-service for all components
Service Discovery	Consul + CoreDNS	Health-aware service registry with automatic failover and load balancing

9.2 Data Platform

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Hot Store (< 24h)	Apache Cassandra 5.x (multi-region active-active)	Sub-millisecond R/W for real-time entity state and threat scoring
Warm Store (90 days)	ClickHouse 24.x (columnar OLAP)	High-speed analytical queries across 90-day rolling event history
Cold Archive (7 years)	Apache Iceberg on S3 / ADLS / GCS	Cost-efficient long-term retention with time-travel query capability

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Graph Database	JanusGraph + Cassandra + Elasticsearch backend	Adversary knowledge graph: 140M+ nodes, 890M+ edges
Search & IOC	Elasticsearch 8.x (ILM-managed)	8.4M+ IOC index; < 5ms query latency; STIX/TAXII normalised schema
Schema Registry	Apache Avro + Confluent Schema Registry	Typed, versioned schemas for all telemetry — schema evolution with backward compat
Cache Layer	Redis Cluster 7.x	In-memory risk score cache and session state for sub-millisecond SOAR decisions

9.3 Security Platform Controls

COMPONENT	TECHNOLOGY / STACK	PURPOSE
Zero-Trust Network	Istio mTLS + SPIFFE/SPIRE identity	Cryptographic identity for every workload; no implicit trust within platform boundary
Supply-Chain Security	Sigstore / Cosign + SBOM (CycloneDX)	Every ACIS build artifact signed; full software bill of materials published per release
Container Security	Falco (runtime) + Trivy (image scanning)	Real-time container syscall monitoring and vulnerability scanning in CI/CD pipeline
Code Security	Semgrep (SAST) + OWASP ZAP (DAST)	Static and dynamic security analysis on every ACIS platform commit
Key Management	HashiCorp Vault + AWS KMS / Azure Key Vault	Envelope encryption with FIPS 140-2 Level 3 HSM-held root keys
Data Encryption	AES-256-GCM (at rest) + TLS 1.3 (in transit)	All ACIS telemetry encrypted at field level at rest and in transit
Network Policy	Kubernetes NetworkPolicy + Cilium eBPF	Layer 3/4 network segmentation enforced at kernel level for all workloads

10. DEPLOYMENT ARCHITECTURE & INTEGRATION ECOSYSTEM

10.1 Deployment Topologies

- ▶ Cloud-Native SaaS: ACIS-managed multi-tenant platform with dedicated data partitioning per tenant. Fastest time-to-value; 48-hour deployment. Ideal for organisations without on-premises infrastructure requirements.
- ▶ Private Cloud / On-Premises: Full ACIS stack deployed within customer-controlled infrastructure via Helm chart deployment. Supports air-gapped environments. Required for organisations with data sovereignty mandates.
- ▶ Hybrid Mesh: ACIS edge processing nodes deployed on-premises feeding anonymised telemetry to cloud-hosted AI, Threat Swarm, and Red Team planes. Optimal for regulated industries (financial services, healthcare, government).

10.2 Integration Ecosystem

ACIS integrates natively with the enterprise security ecosystem via 140+ certified connectors, enabling deployment alongside existing investments without rip-and-replace.

- ▶ SIEM: Microsoft Sentinel, Splunk ES, IBM QRadar, Elastic SIEM — bi-directional alert and event synchronisation; ACIS can operate as primary SIEM or augmentation layer
- ▶ EDR/EPP: CrowdStrike Falcon, SentinelOne, Microsoft Defender for Endpoint, Carbon Black — co-existence mode with telemetry consolidation or full ACIS agent replacement
- ▶ ITSM: ServiceNow, Jira Service Management — automated ticket creation with full ACIS incident context, enrichment, and closure on remediation completion
- ▶ Identity Providers: Okta, Azure Active Directory, Ping Identity, ForgeRock — real-time authentication telemetry ingestion and autonomous response action APIs
- ▶ NGFW: Palo Alto Networks, Fortinet FortiGate, Check Point, Cisco Firepower — dynamic policy push for network isolation and IOC-based blocking
- ▶ Threat Intelligence Platforms: MISP, OpenCTI, Mandiant Advantage, Recorded Future — STIX/TAXII 2.1 feed ingestion and bi-directional IOC sharing
- ▶ Cloud Security: AWS Security Hub, Azure Defender for Cloud, GCP Security Command Center — native cloud security posture integration
- ▶ Vulnerability Management: Tenable.io, Qualys, Rapid7 InsightVM — asset vulnerability context enriching ACIS risk scoring and patch orchestration

11. SECURITY, PRIVACY & COMPLIANCE

11.1 Platform Security Architecture

- ▶ Zero-trust architecture: no implicit trust between any ACIS components; all inter-service communication requires mTLS with SPIFFE-issued certificates and sub-hour rotation.
- ▶ Immutable infrastructure: all ACIS platform components deployed as immutable containers — no SSH access to production nodes, no mutable runtime state, no configuration drift.
- ▶ Supply-chain integrity: all build artifacts signed with Sigstore/Cosign; CycloneDX SBOM published for every release with cryptographic provenance chain.
- ▶ Penetration tested quarterly by CREST-accredited firms; continuous BAS via Layer 5 AI Red Team Simulator operating against the ACIS platform itself.
- ▶ Bug bounty programme with CVSSv4 severity-aligned payouts and coordinated responsible disclosure policy enforced with 90-day resolution SLA.

11.2 Compliance Framework Coverage

FRAMEWORK	ACIS COVERAGE
ISO/IEC 27001:2022	Full Annex A control mapping; Layer 4 auto-generates Annex A.16 incident management reports
SOC 2 Type II	Annual audit with continuous control monitoring via ACIS Compliance Dashboard; reports on request
NIST CSF 2.0	Complete Identify / Protect / Detect / Respond / Recover function coverage across all five layers
GDPR	Article 32 (technical measures), Article 33 (breach notification automation < 72 hours)
ISO/IEC 27701:2019	Privacy Information Management

12. PLATFORM PERFORMANCE SPECIFICATIONS

ATTRIBUTE	SPECIFICATION
Telemetry Ingestion Throughput	10,000,000+ events per second (horizontally scaled, no cap)
AI-SIEM Detection Latency (P99)	< 10 milliseconds from event arrival to risk score emission
CLDM Response Latency (P99)	< 200 milliseconds from threat confirmation to first containment action
ML Inference Latency (P99)	< 5 milliseconds (NVIDIA Triton + TensorRT GPU-accelerated serving)
IOC Fleet Propagation	< 60 seconds from confirmed IOC to fleet-wide blocking
Threat Narrative Generation	< 30 seconds from detection to LLM-generated executive report
Platform Availability SLA	99.999% (< 5 minutes downtime per year; active-active multi-cloud)
Concurrent Monitored Entities	Unlimited — horizontally scaled Kafka + Flink + Cassandra data plane
Attack Simulation Coverage	1,400+ scenarios; 100% MITRE ATT&CK v14 tactic coverage
False Positive Rate	< 0.7% at full production throughput (adversarial training hardened)
Hot Data Retention	24 hours (Apache Cassandra — sub-millisecond access)
Warm Data Retention	90 days (ClickHouse OLAP — < 100ms analytical query)
Cold Data Retention	7 years (Apache Iceberg on object store — time-travel capable)
Supported Endpoint Platforms	Windows 10+, macOS 12+, Linux (kernel 4.18+), Kubernetes, iOS, Android
Log Source Parsers	850+ native; 1,200+ via OpenTelemetry collector
Security Integrations	140+ certified connectors across EDR, NGFW, IdP, ITSM, SIEM, and cloud

13. CONCLUSION

The modern cybersecurity challenge cannot be solved by adding more analysts, more tools, or more rules to an architecture that was designed for a slower, simpler threat landscape. The fundamental problem is structural: human reaction time has become the limiting constraint in a world where adversaries operate at machine speed.

ACIS resolves this structural failure through a five-layer, bio-inspired autonomous defence architecture that eliminates human latency from every critical phase of the security operations lifecycle. The AI-Powered SIEM provides continuous intelligent detection at scale. The Autonomous SOAR responds and contains within 200 milliseconds. The Threat Intelligence Swarm builds collective immunological memory across the entire fleet. The Self-Healing and Deception Engine autonomously remediates and misdirects adversaries. The AI Red Team Simulator continuously validates that every control works — not once a year, but every hour of every day.

Together, these five layers form the Closed-Loop Defence Mode: a continuously operating autonomous immune system that detects, responds, heals, learns, and validates — at a speed and scale no human team can match. ACIS does not replace security professionals; it frees them from the mechanical, latency-bound work of detection and containment so they can focus on strategic threat intelligence, architecture, and adversary pursuit.

ACIS — FIVE LAYERS. ONE IMMUNE SYSTEM.

- Layer 1 — AI-Powered SIEM: 10M+ events/sec · < 10ms detection · 8.4M+ IOC signatures · 850+ log sources
- Layer 2 — Autonomous SOAR: < 200ms response · 240+ playbooks · 140+ integrations · atomic rollback
- Layer 3 — Threat Intelligence Swarm: federated learning · 140M-node adversary graph · < 60s IOC propagation
- Layer 4 — Self-Healing & Deception Engine: automated remediation · patch orchestration · active deception
- Layer 5 — AI Red Team Simulator: 1,400+ scenarios · 24/7 continuous · 100% ATT&CK tactic coverage
- Platform SLA: 99.999% availability · Multi-cloud active-active · ISO 27001 · SOC 2 Type II · GDPR · PCI-DSS

ACIS — Autonomous Cyber Immune System

© 2025 · Netcradus Pvt. Ltd. · Classification: Confidential · All Rights Reserved